College Decision Analysis for Minnie Bui

I. Student Profile

Minnie Bui is a 17-year-old Vietnamese international student currently completing her final year of high school education. She presents a strong academic profile, with a cumulative GPA of 3.7/4.0 and a combined SAT score of 1380/1600, including a 700 in Math and 680 in Reading. Her primary academic interest lies in actuarial science, a quantitative discipline combining mathematics, statistics, and financial theory, with career aspirations in insurance, risk management, and financial consulting.

As an international student relocating independently to the United States, Minnie faces significant financial constraints. Affordability is the central decision factor: she requires institutions that offer strong financial aid packages, a low net cost of attendance of $25000 per year, and accessible co-op or work-study opportunities to supplement her funding. She prefers a medium-sized institution with 5,000 to 15,000 undergraduate students, located in an urban or suburban setting, especially on either the East or West Coast, with proximity to the finance and insurance industry. She values a collaborative, academically rigorous environment with strong international student support services. Her sister is currently studying in Indiana, which makes Midwest institutions an additional point, though not necessarily important for geographic preference.

II. Data Overview

This analysis joins two institutional datasets on the school identifier (*OPEID6*). The one-to-many relationship between the Institutional Data and Field of Study tables reflects the fact that each institution offers multiple programs. This relationship allows the dashboard to simultaneously display institutional characteristics, such as cost and selectivity, alongside program-specific outcomes, such as graduate earnings and student debt, enabling an evaluation of each school that neither dataset could support on its own.

Institutional Data

The Institutional Data file provides school-level characteristics for post-secondary institutions across the United States and its territories. Key fields used in this analysis include admission rate (*ADM_RATE*), total annual cost of attendance (*COSTT4A*), net price after financial aid for public institutions (*NPT4_PUB*) and private institutions (*NPT4_PRIV*), SAT mathematics and reading 25th percentile scores (*SATMT25*, *SATVR25*), total undergraduate enrollment (*UGDS*), campus locale classification (*LOCALE*), state abbreviation (*STABBR*). The locale field was grouped into four categories: City, Suburban, Town, and Rural, to support geographic preference filtering. A calculated net cost field was derived by combining the public and private net price fields using coalesce logic, selecting the populated value for each institution.

Field of Study Data

The Field of Study Data file provides program-level outcome metrics for specific fields of study within each institution, identified by six-digit CIP classification codes. Key fields used in this analysis include median graduate earnings one year post-enrollment (*EARN_MDN_HI_1YR*), median graduate earnings two years post-enrollment (*EARN_MDN_HI_2YR*), and median cumulative student debt at program completion (*DEBT_ALL_STGP_EVAL_MDN*). Each row in this file represents one program at one institution, meaning a single institution may appear multiple times across different CIP codes. Aggregation to the institution level was performed using average functions in Tableau to produce comparable school-level metrics.

Calculated Fields

- **Calculated Net Cost**: combines public and private net price data into a single measure
- **SAT Eligibility**: classifies each institution as Eligible, Not Eligible, or SAT Not Required based on a comparison between the user-entered SAT parameter values and the institution's reported 25th percentile scores
- **Within Budget**: filters institutions dynamically based on the user's entered maximum annual budget parameter. A Fit Score field aggregates four weighted criteria, admission accessibility, affordability, graduate earnings, and SAT eligibility, into a single composite score out of 100
- **Fit Label** field translates the numeric fit score into four descriptive tiers: Best Fit (85–100), Target (65–84), Reach (50–64), and Not Ideal (below 50)

Filtering

Schools were filtered throughout this analysis to retain only those meeting all of the following criteria simultaneously. First, the institution must offer a bachelor's degree in Mathematics or a related quantitative field, confirmed by the **CIP27BACHL** flag, corresponding to CIP classification codes in the **27XX** series covering Mathematics, Applied Mathematics, Statistics, and Management Sciences. Second, the institution must offer a bachelor's degree in Business or a related field, confirmed by the **CIP52BACHL** flag, corresponding to CIP classification codes in the **52XX** series covering Business Administration, Finance, Accounting, and Quantitative Methods.

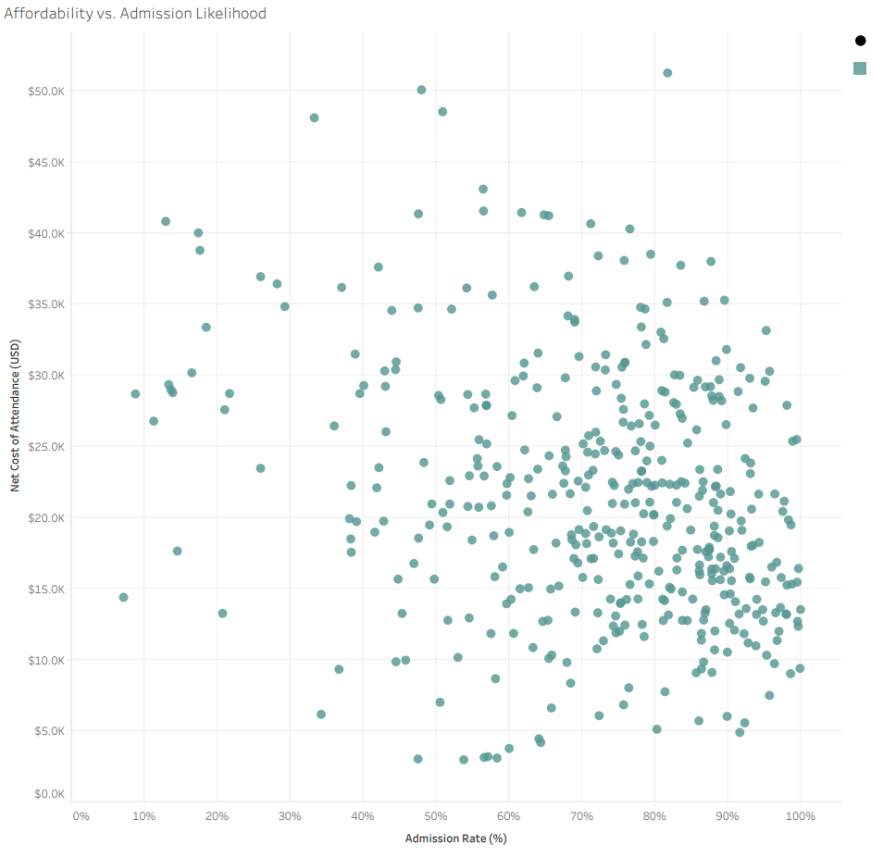
Limitations

Several data quality considerations affect the interpretation of results. Net cost figures reflect averages across all enrolled students and do not specifically represent international student costs, which are typically higher at public institutions due to out-of-state tuition structures. SAT score data are missing for a substantial number of institutions, either because the school is test-optional or test-blind, or because it did not report scores to the federal data collection. These institutions are classified as SAT Not Required and retained in the analysis with a visual distinction from confirmed eligible schools. Graduate earnings figures reflect median outcomes across all program completers and do not isolate mathematics or actuarial science graduates specifically. Enrollment figures and cost data reflect the most recent available reporting year in the dataset. They may not

reflect current institutional pricing, particularly given post-pandemic tuition adjustments at many institutions.

III. Chart Analysis for College Decision

1. Affordability vs Admission Likelihood



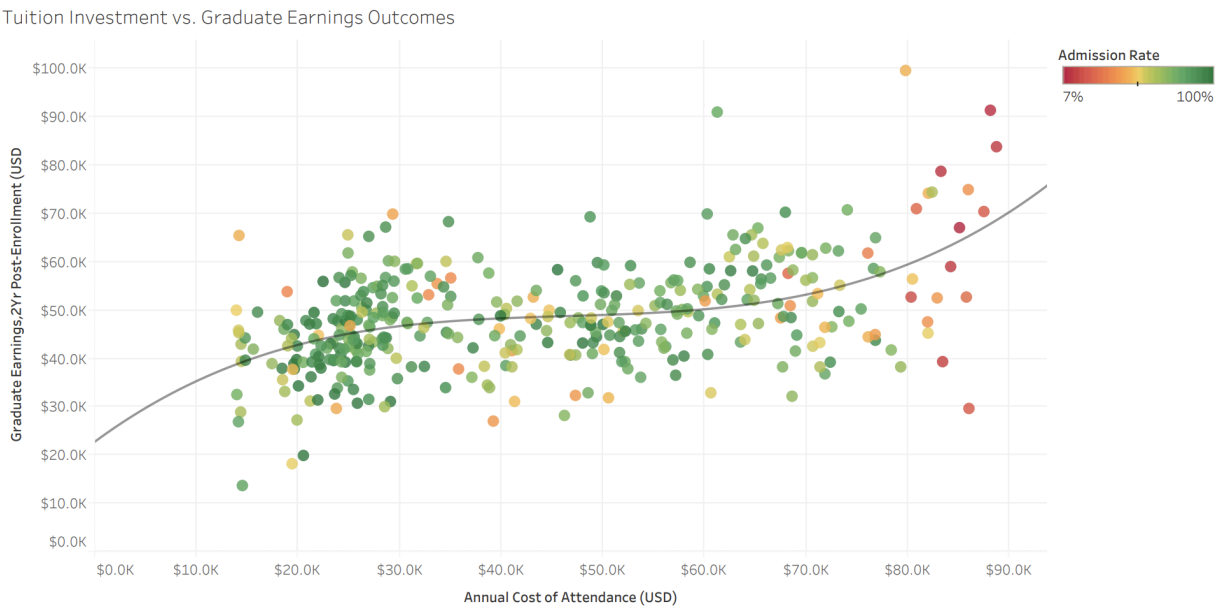
This scatter plot is the primary decision chart. Each dot represents one institution. The horizontal axis shows how selective the school is — schools on the left are harder to get into, and schools on the right admit a higher proportion of applicants. The vertical axis shows the estimated annual cost after financial aid is applied, which is closer to what the student would actually pay than the advertised sticker price.

Teal circle dots are institutions where the applicant's SAT scores meet or exceed the school's 25th percentile threshold. These are schools where the applicant is academically competitive. The chart is filtered to show only institutions within the user's stated budget, so every visible dot is both financially accessible and academically realistic.

The most valuable region of this chart is the bottom-right quadrant, schools with high admission rates and low net cost. These represent the strongest combination of financial accessibility and admissions realism. Schools in the upper-left quadrant are expensive and selective; they may offer prestige but carry significant financial and admissions risk for an international student without guaranteed external funding.

The tooltip visible in the dashboard screenshot shows Augusta University with a 77% admission rate, a net cost of $18.8k, a combined SAT score of 1060, and confirmed SAT eligibility. This strong candidate scores 95 on the fit scoring model.

2. Tuition Investment vs Graduate Earnings

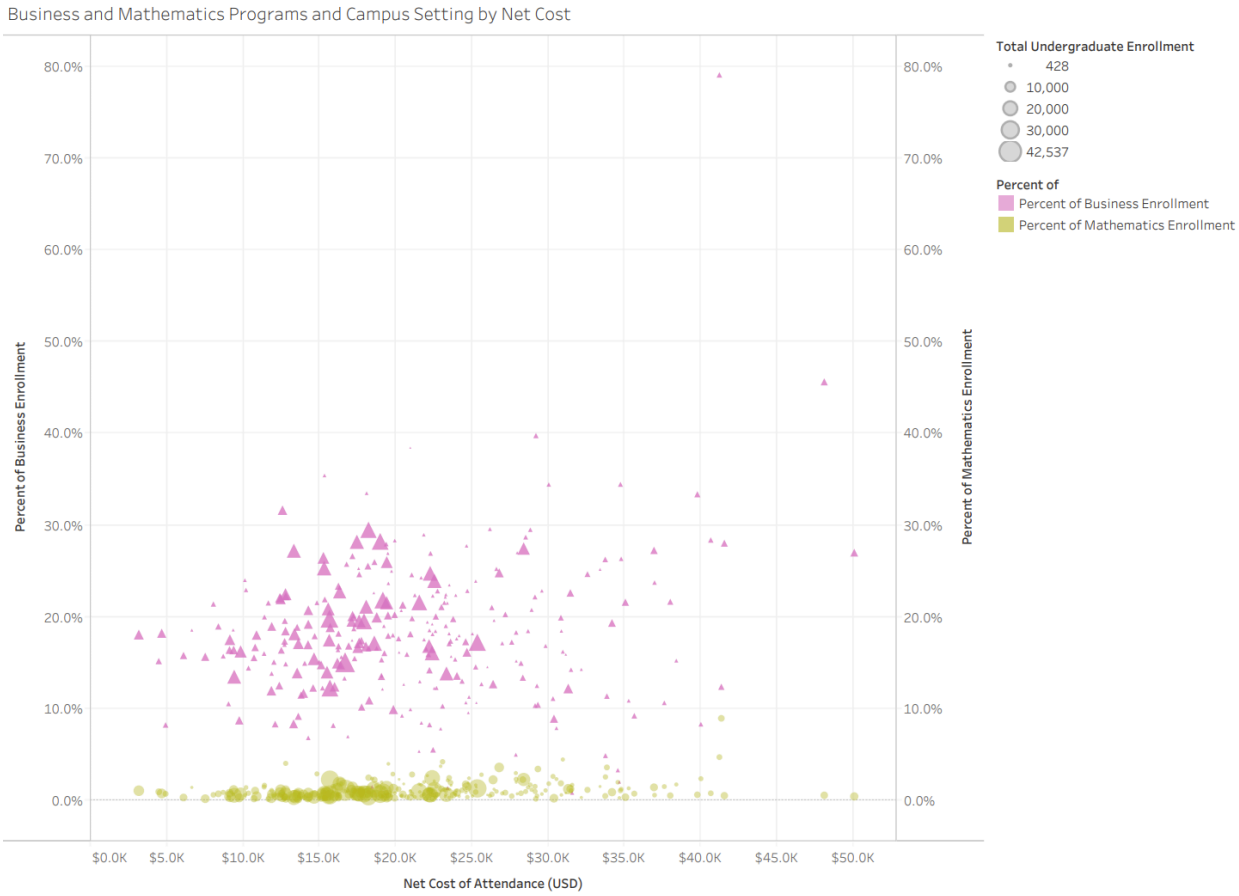


This scatter plot examines the financial return on a college education. The horizontal axis shows the average annual sticker price of each institution, while the vertical axis shows median graduate earnings two years after enrollment. Each dot is colored by admission rate: red dots are highly selective institutions, and green dots are more accessible.

The logarithmic trend line running through the data is the most important analytical element on this chart. It shows that the relationship between cost and earnings is not proportional (R^2=0.21), paying significantly more does not produce proportionally higher earnings. The curve rises steeply at the low-cost end, meaning that moving from a $10k school to a $20k school produces a noticeable improvement in expected earnings. However, the curve flattens considerably above $40k annual cost, meaning the earnings premium from attending a $70k school versus a $40k school is minimal. For an international student managing financial constraints, this finding strongly supports targeting mid-range-cost institutions rather than high-prestige, expensive schools.

The ideal schools on this chart are green dots appearing in the upper-left region, with accessible admission rates, relatively low sticker prices, and graduate earnings above $50k. The red dots in the lower-right are the most financially questionable choices: expensive, selective, and yielding moderate earnings outcomes.

3. Business and Mathematics Programs and Campus Setting by Net Cost



This dual-axis chart shows how strongly different institutions invest in mathematics and business programs relative to their net cost, with dot size encoding total undergraduate enrollment. Blue triangles represent the percentage of students enrolled in business programs, while purple circles represent the percentage enrolled in mathematics programs. The x-axis shows net cost after aid, placing cheaper schools on the left and more expensive schools on the right.

For an actuarial science applicant, the most relevant dot is the green circle: the percentage of students enrolled in the mathematics program. Schools where green circles appear high on the right-hand side of the right-side axis, relative to their cost position on the x-axis, represent institutions with a genuine mathematical academic culture at a reasonable price. A green circle at $15k net cost and 3% mathematics enrollment is significantly more valuable than one at $40k net cost and the same enrollment percentage.

The size of each dot shows total undergraduate enrollment, allowing the viewer to distinguish a small liberal arts college from a large state university at a glance. Minnie's preference for medium-sized institutions of 5,000 to 15,000 students means she should focus on medium-sized dots rather than the largest circles.

The dual-axis means the left and right percentage scales are different. Business enrollment reaches 80% while mathematics enrollment peaks at around 9%. This reflects the reality that business is the most common undergraduate major in the United States while mathematics

programs enroll a much smaller share of students. A school with 3% enrollment in mathematics is actually a strong mathematics institution by national standards.

4. Interactive Filtering Dashboard

All four charts are connected through Tableau's action filters. Clicking any dot on any chart highlights that institution across all other charts simultaneously. This means that if Minnie identifies a school on the Affordability chart, she can immediately see where that school falls on the earnings curve, what its program enrollment looks like, and what its fit score is in the ranked list, without having to search manually. Adjusting any parameter in the Filter Fields panel cascades across all charts within approximately one second, making the tool genuinely interactive rather than static.

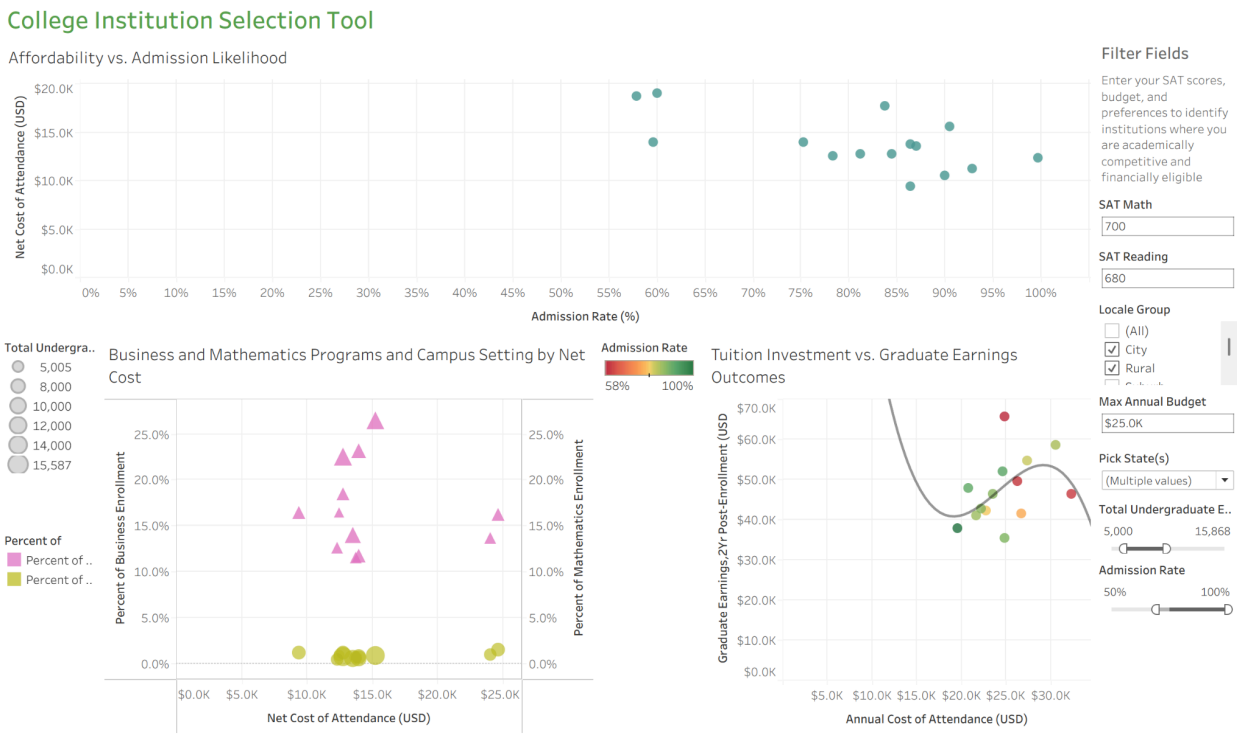
When hovering over a single dot on the Affordability vs Admission chart, the tooltip shows the institution name, admission rate, net cost, SAT score, SAT eligibility status, fit score, enrollment size, sticker price, completion rate, median debt, and graduate earnings, all in a single panel. This eliminates the need to cross-reference multiple data sources during the decision-making process.

IV. Conclusion

Institution List

		Fit Score	Total Undergraduate Enrollment	Annual Cost of Attendance ..	4Yr Completion Rate	Debt All Stgp Eval Mdn	Earn Mdn Hi 2Yr
Augusta University	GA	95	5,613	$23.5K		$22.6K	$46.4K
Coastal Caroli University	SC	95	10,377	$26.7K		$24.0K	$41.5K
College of Charleston	SC	90	10,558	$32.4K		$23.3K	$46.4K
Embry-Riddle Aeroutical ..	FL	85	9,139	$24.9K	32.04%	$28.1K	$65.6K
Lamar University	TX	95	8,150	$22.3K	39.87%	$25.8K	$42.7K
McNeese State University	LA	95	5,005	$22.8K		$23.4K	$42.2K
Middle Georgia State Uni..	GA	90	6,574	$19.6K		$23.0K	$37.8K
Texas A&M University-Sa..	TX	90	6,599	$21.7K	30.19%	$19.3K	
University of Akron Main ..	OH	85	9,052	$26.3K	14.63%	$24.8K	$49.6K
University of Houston-Cle..	TX	100	6,078	$24.6K	52.63%	$20.8K	$52.0K
University of Houston-Do..	TX	95	12,555	$20.7K		$21.9K	$47.9K
University of Louisia at La..	LA	90	12,428	$24.9K	82.98%	$23.2K	$35.4K
University of Massachuse..	MA	100	11,512	$30.5K		$24.9K	$58.5K
Wayne State University	MI	100	15,587	$27.4K		$23.1K	$54.7K
Youngstown State Univer..	OH	95	7,961	$21.7K	63.71%	$22.6K	$40.9K

Fit Label
- Safety
- Target



The University of Massachusetts Boston is the top coastal recommendation, with a fit score of 100, within budget at $24.9k, $58.5k earnings, and a city campus in Boston, one of the strongest actuarial employment markets in the country. The enrollment of 11,512 is squarely in Minnie's preferred range.

University of Houston Clear Lake is a new entry worth noting, with a fit score of 100, a $24.6k net cost within budget, $52k in earnings, and 6,078 enrollment, which is the ideal medium size. Houston is a major hub for the insurance and energy sectors, with strong actuarial employment. However, Texas is a strong backup if coastal options fall through.

Wayne State is $27.4k net cost, which is slightly over Minnie's $25k budget, but the score of 100 and $54.7k earnings make it worth applying with a financial aid negotiation request.